

Homework: SVM Classification

Dataset

Use the **Titanic Dataset**.

Objective

Build an SVM classifier to predict whether a passenger survived the Titanic disaster.

Requirements

1. Download the Titanic dataset from Kaggle.
2. Perform complete EDA:
 - Missing value analysis
 - Class imbalance analysis
 - Correlation analysis
 - Detect categorical and numerical features
3. Preprocess the dataset:
 - Handle missing values appropriately (justify your method).
 - Encode categorical variables.
 - Drop irrelevant features only if you can justify why.
 - Scale numerical features.
4. Create **two different feature sets**:
 - Feature Set A: Original cleaned features.
 - Feature Set B: Feature engineered dataset (e.g., FamilySize, IsAlone, Title extracted from Name, Age Groups, Fare Groups, etc.).
5. Train SVM models using:
 - Linear Kernel
 - Polynomial Kernel
 - RBF Kernel
6. For each kernel:
 - Show Training Accuracy
 - Show Testing Accuracy
 - Plot the Confusion Matrix
 - Print the Classification Report
7. Compare all models and identify:

- Which kernel overfits?
 - Which kernel generalizes best?
 - Explain why.
8. Discuss why feature scaling is essential for SVM. Demonstrate this by training an SVM **without scaling** and compare the results with the scaled version.
9. Write a **300-word analysis** explaining:
- Why SVM struggled or succeeded on this dataset.
 - Which engineered features contributed most to performance.
 - Whether SVM is the best choice for this problem.